IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2021

POWER ELECTRONICS

Thursday, 6 May 10:00 am

Time allowed: hours

There are THREE questions on this paper.

Answer questions.

Any special instructions for invigilators and information for candidates are on page 1.

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Information for Candidates

Permittivity of free space: $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m

Permeability of free space: $\mu_0 = 4\pi \times 10^{-7}$ H/m

Switch-Mode Power Supplies

Voltage ratio equations for buck SMPS $\frac{V_O}{V_I} = \delta$ or $\frac{V_O}{V_I} = \frac{1}{1+\frac{2fL\,I_O}{V_I\delta^2}}$

Voltage ratio equations for boost SMPS $\frac{V_O}{V_I} = \frac{1}{1-\delta}$ or $\frac{V_O}{V_I} = \frac{1}{1-\frac{V_I\delta^2}{2fL\,I_I}}$

Three-Phase Systems

Line Voltages and Current

Star $V_L = \sqrt{3}\,V_P;\ \ I_L = I_P$ Delta $V_L = V_P;\ \ I_L = \sqrt{3}\,I_P$

Real Power $P_{3\phi} = 3\,V_P I_P cos(\phi) = \sqrt{3}\ V_L I_L cos(\phi)$

Induction Machines

$\omega_S = {\omega_E}/{P}$; $\quad s = \frac{\omega_S - \omega_R}{\omega_S}$; $\quad T_{EM} = \frac{3\,I'^2_R\,R'_R}{\omega_R}\left(\frac{1-s}{s}\right)$

Power Flow in Lines and Cables

Cable Parameters (expressed per metre)

$R'_{LF} = \frac{1}{\sigma_C \pi\, r_C^2} + \frac{1}{\sigma_C 2\pi\, r_O\, t_O}$ $\quad L' = \frac{\mu}{2\pi} ln\left({r_O}/{r_C}\right)$ $\quad C' = \frac{2\pi\varepsilon_0\varepsilon_{RI}}{ln\left({r_O}/{r_C}\right)}$ $\quad G' = \frac{2\pi\sigma_I}{ln\left({r_O}/{r_C}\right)}$

OHL Parameters in approximate form (expressed per metre)

$R'_{LF} = \frac{2}{\sigma_C \pi\, r_C^2}$ $\quad L' = \frac{\mu}{\pi} ln\left({d}/{r_C}\right)$ $\quad C' = \frac{\pi\varepsilon_0\varepsilon_{RI}}{ln\left({d}/{r_C}\right)}$ $\quad G' = \frac{\pi\sigma_I}{ln\left({d}/{r_C}\right)}$

Power Flow (full form)

$$P_S = \frac{{V_S}^2}{Z_{SR}}\cos(\theta) - \frac{V_R V_S}{Z_{SR}}\cos(\theta+\delta) \qquad Q_S = \frac{{V_S}^2}{Z_{SR}}\sin(\theta) - \frac{V_R V_S}{Z_{SR}}\sin(\theta+\delta)$$

Voltage Drop (approximate form) $\Delta V = |V_S| - |V_R| \approx \frac{R\,P_S + X\,Q_S}{|V_S|}$

1. This question covers several topics and all parts should be attempted.

a) Consider an overhead line used for a single-phase AC supply. The line has a series impedance of $j30\ \Omega$, that is, it is purely inductive.

i) Calculate the real and reactive powers at the sending end if the sending end voltage is 220 kV and the receiving end voltage is 215 kV lagging the sending end by 6.0°. [6]

ii) Calculate the maximum real power this line can transfer when operated at 220 kV at both ends if the only limit on power transfer is the angle criteria. [4]

iii) The line has a maximum current carrying capability of 3,500 A. For operation of this line under this limit, and with both ends at 220 kV, calculate the load angle at which the line would operate and the power transfer this creates. [6]

b) Consider a three-phase AC supply with a line voltage of 11 kV.

i) Three impedances of $(6.0 - j\ 3.5)\ \Omega$ are formed into a star connection and connected to the supply. Calculate the phase current and line currents that flow. Provide the magnitudes of both currents and the angle of a phase current relative to a phase voltage at 0°. [4]

ii) Calculate the real and reactive powers drawn by the star-connected impedances. [4]

iii) One of the three impedances is found to be $(5.5 - j\ 3.0)\ \Omega$ (the other two impedance remain at $(6.0 - j\ 3.5)\ \Omega$. Calculate the magnitude of the current that flows in the neutral in this situation. [4]

c) Consider an induction machine.

i) Explain what determines the rate of change of flux linkage of the rotor bars of an induction machine. [4]

ii) Explain what power loss the magnetising resistance represents. [4]

iii) Explain what is meant by flux leakage, how it is represented in an equivalent circuit and why it does not directly impact the efficiency of the machine. [4]

2. This question concerns **buck** switch-mode power supplies (SMPS)

a) i) The voltage transfer ratio of buck SMPS is often quoted as $\frac{V_O}{V_I} = \delta$ where δ is the duty-cycle of the switch. Explain the basis on which this equation is derived (you need not reproduce the derivation itself) and through that explain why calculation of output voltage using this expression is not completely accurate in practice. [5]

ii) Derive an expression for the output voltage of a Buck SMPS in continuous conduction that accounts for voltage drops within the circuit. [5]

b) A **buck** SMPS is to be designed to convert 24.0 V to 5.0 V with output currents of up to 8.0 A.

The on-state resistance of the chosen MOSFET is 0.14 Ω. The turn-on and turn-off energy losses of the MOSFET are 1.5 μJ and 1 μJ when operated at 1 A and **scale linearly with current**. The MOSFET is mounted on a heat-sink which gives a total thermal impedance between junction and air of $R_{ThJA} = 12.5\ °C/W$.

The SMPS is required to work in ambient temperatures of up to 45°C and the junction temperature is to be no greater than 105°C.

i) Calculate the maximum switching frequency at which the MOSFET can be operated in this design. You may assume the SMPS will be in continuous conduction mode when operating at maximum current. [6]

ii) The capacitor suggested for the design has a capacitance of $C = 750\ \mu F$ and effective series resistance of $R_{ESR} = 35\ m\Omega$. The output voltage ripple is not to exceed $\Delta v_O = 10\ mV$. Express the total output voltage ripple as the simple sum of terms due to the capacitance and effective series resistance and use this to find the maximum inductor current ripple that can be allowed. [4]

iii) Calculate the value of inductor that should be chosen. [4]

iv) Assume the inductor has a resistance of $R_L = 0.05\ \Omega$ and the diode has a forward voltage drop of $V_D = 0.8\ V$. Calculate the efficiency of the SMPS at full load. [4]

v) It is suggested that to raise the efficiency of this SMPS, synchronous rectification could be used with a second MOSFET of the same characteristics as the main MOSFET. How effective would that be? You may assume that the second, rectifier, MOSFET incurs conduction power loss but no switching power loss. [2]

3. An induction machine which is a 3-phase, 2 pole-pair, star-connected machine is connected to a 50 Hz supply with a phase-voltage of 230 V. The parameters of the equivalent circuit, referred to the stator, are:

Stator resistance,	$R_S = 0.46\ \Omega$;
Stator leakage reactance,	$X_S = 0.51\ \Omega$;
Magnetising resistance,	$R_M = 500\ \Omega$;
Magnetising reactance,	$X_M = 60\ \Omega$;
Referred rotor leakage reactance,	$X'_R = 0.51\ \Omega$;
Referred rotor resistance,	$R'_R = 0.60\ \Omega$.

The machine is operated with a slip of $s = 0.02$.

i) Give an expression for the total input impedance of the machine and calculate its value. [5]

ii) Calculate the magnitude of the rotor current. [5]

iii) Calculate the electromagnetic torque developed. [5]

iv) An approximate expression for funding torque, for small values of slip s, is $T \approx \frac{3\,|V_{Th}|^2}{\omega_S\ R'_R}\ s$ where V_{Th} is the Thévenin equivalent voltage found by removing the referred rotor impedance from the circuit and taking the Thévenin equivalent voltage at that position in the circuit. Compare the torque found from this approximation to the torque found in (iii). [5]

v) The machine is redesigned with thicker conductors in the rotor which halve the referred rotor resistance. Using the approximation in (iv) to estimate the slip at which the machine would operate to develop the same torque as in (iv). [3]

vi) For the condition in (v), calculate the rotor current. Hint: the approximation in (iv) neglects the Thevenin impedance and that approximation can be used here. [2]

vii) Comment on how the power losses in the rotor resistance, stator resistance and magnetising power loss, and the efficiency of the new machine would be expected to compare to the original machine. There is no need to calculate values. [5]